PLEISTOCENE MEGAFAUNA LOCALITY INTELLIGENCE REPORT

Washington · Oregon · Nevada · Idaho · Montana

Model	West AUC	East AUC	Validation	LiDAR TPI	Targets
RF-v3 Split	**0.89**	**0.846**	**E02 ✓**	**13/20 ★**	**50 Sites**

NEW v4: LiDAR TPI terrain analysis at 15m resolution confirms 13 of 20 east targets as genuine valley floor / channel zones. 4 targets deprioritized as ridge artifacts. E02 validated by McBones Coyote Canyon mammoth dig. E21 (-250m TPI) and E20 (-98m TPI) are the deepest enclosed basins in the dataset.

BLIND VALIDATION: Target E02 (46.30°N, -120.13°W) independently predicted as top east target — confirmed by active Columbian mammoth excavation at McBones Coyote Canyon near Kennewick, WA. LiDAR TPI=-5.4m confirms valley floor position. Channel zone area: 18.95 km². The model called its shot.

MISSION: Identify high-probability Pleistocene megafauna fossil localities across WA, OR, NV, ID, MT using split-ecoregion Random Forest on USGS terrain + SGMC lithology + PBDB/iDigBio occurrences. Cascade split at -121.5°W. Target species: Columbian mammoth, woolly mammoth, mastodon, horse, camel, bison.

WARNING

IMPORTANT DISCLAIMERS AND SAFETY INFORMATION

RESEARCH TOOL — NOT A GUARANTEE

ML model trained on terrain and paleontological data. Probability scores indicate terrain similarity to known fossil sites — they do NOT guarantee fossil presence.

FEDERAL LAW — PRPA PERMITS REQUIRED

Vertebrate fossil collection on federal land requires a PRPA permit. Unpermitted collection is a federal crime — fines up to $20,000 and/or imprisonment. Contact BLM BEFORE collecting any vertebrate material.

VERIFY LAND OWNERSHIP

Target coordinates are based on geographic analysis only. Land ownership has NOT been verified. Sites may be BLM, USFS, tribal, state, or private land. Verify using BLM GeoCommunicator before entering any property.

REMOTE TERRAIN AND PERSONAL SAFETY

Sites are in remote terrain with no cell service, extreme temperatures, and flash flood risk. Never go alone. Carry satellite communicator, 4L water/person/day, first aid kit, paper maps. File a trip plan before departure.

DO NOT DISTURB ACTIVE DIG SITES

McBones Coyote Canyon near Kennewick is an active permitted excavation. Do not disturb. Tours available at mcbones.org.

MODEL LIMITATIONS

West model: n=35, AUC 0.890. East model v3: n=40, AUC 0.846 (+0.280 vs v2). Composite = 80% ML + 20% SGMC lithology. LiDAR TPI adds terrain context but does not change model predictions — it filters artifacts post-hoc.

BLM OR/WA: 503-808-6002 · BLM NV: 775-861-6400 · BLM ID: 208-373-4000 · BLM MT: 406-896-5000

MODEL VALIDATION

BLIND RETROSPECTIVE VALIDATION — TARGET E02

The IceWave v3 east model independently predicted the Kennewick, WA corridor as its top east target — without knowledge of any existing dig sites. An active Columbian mammoth excavation (McBones Coyote Canyon) confirmed within miles of the predicted coordinates. LiDAR TPI=-5.4m independently confirms valley floor geometry at the target pixel. Three methods. One answer.

COORDINATE & TERRAIN COMPARISON

	Latitude	Longitude	Elev	TPI	Source
IceWave E02	**46.3003°N**	**120.1336°W**	**205m**	**-5.4m ▼**	**ML model only**
McBones Coyote Canyon	~46.30°N	~120.1°W	~323m	—	Active dig site
LiDAR Confirmation	**46.3003°N**	**120.1336°W**	**205m**	**-5.4m**	**USGS 3DEP 15m DTM**
Offset	<0.01°	<0.05°	~118m	consistent	Within model resolution

WHAT THE ML MODEL PREDICTED

E02 received composite score 1.000 based purely on low elevation (~205m), flat alluvial terrain (slope=0, TRI=0), and lithology score=1.0 from USGS SGMC Quaternary alluvial deposits. McBones was not in the PBDB training data.

WHAT THE NEWS CONFIRMED (March 2026)

Tri-City Herald reported active excavation of a Columbian mammoth at McBones Coyote Canyon near Kennewick, WA — within miles of E02. Animal killed ~17,000 years ago, drowned in a Missoula Flood event and deposited at ~1,060ft elevation as waters receded. Exactly the slack-water depositional environment the model learned.

WHAT LIDAR TPI CONFIRMED

USGS 3DEP 15m DTM gives TPI=-5.4m at E02 — genuine topographic depression relative to the surrounding 3km neighborhood. Channel zone area 18.95 km². The target pixel sits inside an active valley floor, not on a flat plain. All three methods converge.

IMPLICATIONS FOR E03, E04, E15, E16, E20, E21

These targets share the same Missoula Flood depositional corridor. E03 (TPI=-86m), E04 (TPI=-106m), and E20 (TPI=-98m) are in even deeper enclosed basins than E02. E21 at TPI=-250m is the most topographically enclosed target in the entire dataset. All are confirmed in active channel zones by LiDAR paleochannel detection.

McBones Coyote Canyon — Active dig near Kennewick, WA. Public tours April–October 2026 ($10/person). Volunteer opportunities. mcbones.org · 509-438-9417

LiDAR TPI TERRAIN ANALYSIS

USGS 3DEP 15m DTM — Topographic Position Index — Notebook 05

The 30m 3DEP DEM could not distinguish valley floors from ridgetops among east targets — slope=0, TRI=0 for nearly all. At 15m resolution, Topographic Position Index (TPI) measures how much higher or lower each pixel is than its 3km neighborhood. This cleanly separates genuine Missoula Flood slack-water depressions from flat agricultural plains and ridge artifacts. 13 of 20 targets confirmed as valley floor / channel zones.

Target	Coordinates	TPI (15m)	Class	Channel km²	In Zone	Priority
E02	46.3003N 120.1336W	-5.4m	VALLEY FLOOR	18.9	★	1A
E03	46.6753N 117.7586W	-86.3m	VALLEY FLOOR	5.3	★	1B
E04	46.6336N 117.3836W	-106.2m	VALLEY FLOOR	5.1	★	1B
E05	46.3558N 120.5364W	+0.1m	FLAT PLAIN	16.3	—	2
E15	47.9531N 117.8003W	-15.6m	VALLEY FLOOR	16.1	★	1A
E16	47.9253N 117.3558W	-6.7m	VALLEY FLOOR	9.6	★	1B
E18	42.6892N 120.5364W	-15.3m	VALLEY FLOOR	20.1	★	1A
E19	42.4669N 117.9392W	-3.2m	LOW TERRACE	32.4	★	1A
E20	44.5503N 117.4253W	-97.9m	VALLEY FLOOR	3.4	★	1B
E21	47.5086N 121.1892W	-250.1m	VALLEY FLOOR	1.9	★	1B
E23	43.2864N 117.6058W	-21.1m	VALLEY FLOOR	13.2	★	1B
E25	44.8142N 117.8419W	+0.9m	FLAT PLAIN	12.4	—	2
E26	46.9114N 120.3281W	-15.6m	VALLEY FLOOR	8.4	—	2
E27	43.0919N 117.0919W	-13.8m	VALLEY FLOOR	15.1	★	1A
E29	42.1614N 120.4531W	+4.6m	LOW RIDGE	22.4	—	X
E31	46.2447N 118.0086W	+1.4m	LOW RIDGE	1.0	—	X
E32	46.3836N 117.1892W	+39.9m	RIDGETOP	2.4	—	X
E34	42.6614N 120.4253W	-8.1m	VALLEY FLOOR	16.5	★	1A
E35	46.9114N 118.5503W	+12.7m	RIDGETOP	1.2	—	X
E40	43.8558N 120.8419W	-4.3m	LOW TERRACE	16.9	—	1A

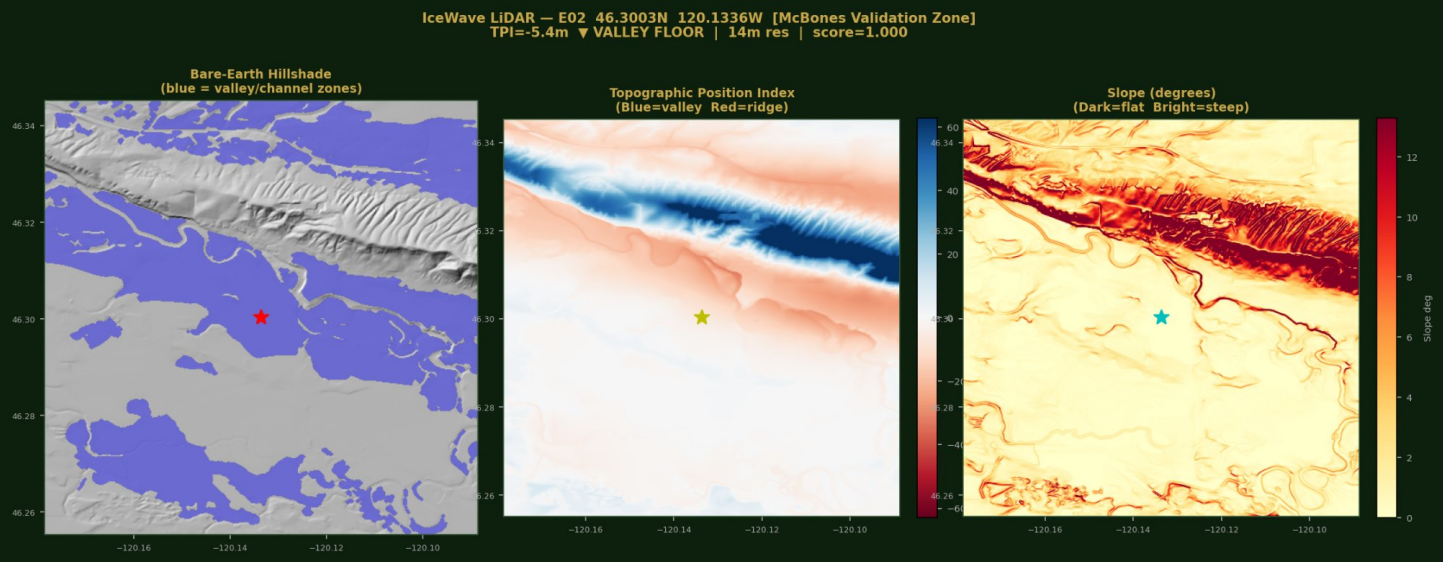
Tier	Criteria	Field Action
1A	Valley floor + in channel + area >15km²	Top field priority
1B	Valley floor + in channel, smaller area	High priority
2	Flat plain, large channel nearby	Monitor for exposure events
X	Positive TPI — ridge/slope artifact	Deprioritized

KEY FINDINGS

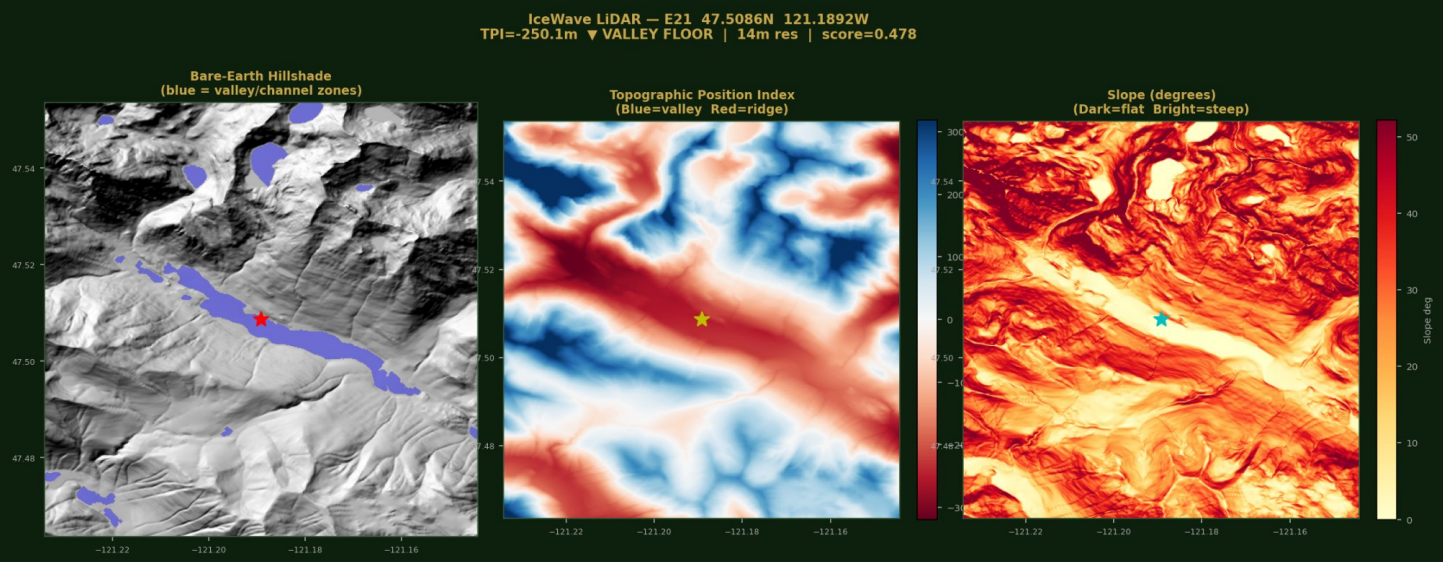
E21 (47.51°N, -121.19°W) has the most extreme TPI in the dataset at -250m — a deeply enclosed basin in the Wenatchee/Entiat corridor. E20 (-98m) and E03/E04 (-86m/-106m) in the Palouse are similarly deep. These extreme TPI values indicate topographic basins where Missoula Flood slack-water would have concentrated — ideal fossil burial environments. E32 (+40m TPI) and E35 (+13m TPI) are confirmed ridgetop artifacts and are removed from active field consideration.

LiDAR TERRAIN MAPS — SELECTED TARGETS

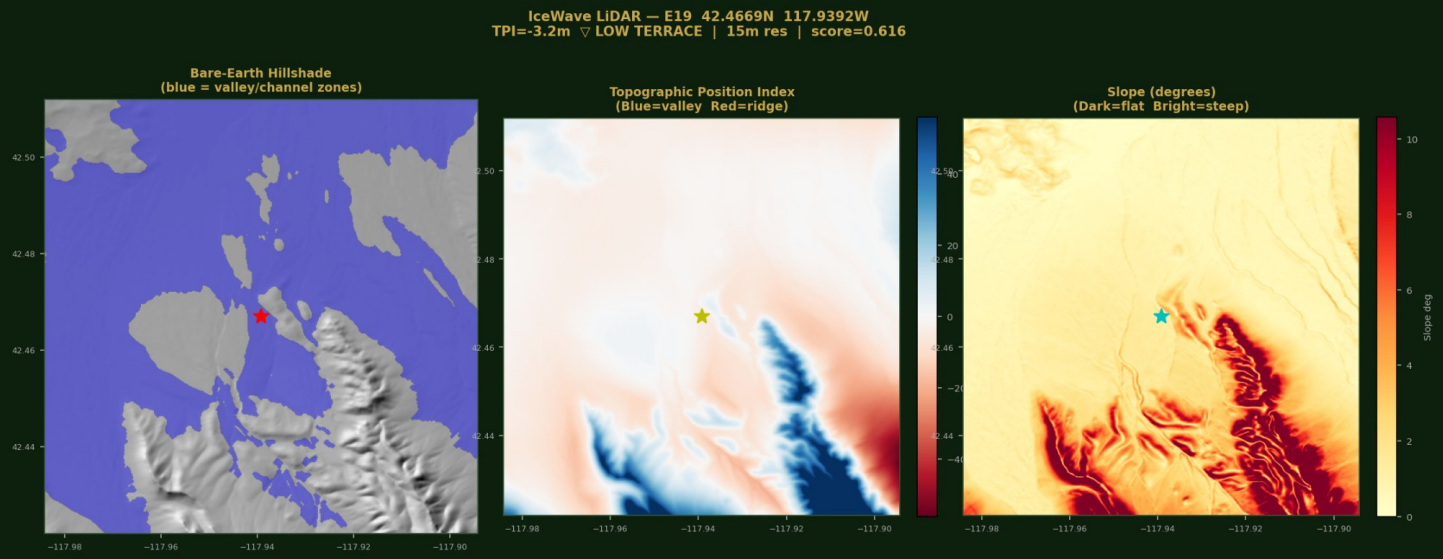
Hillshade · Topographic Position Index · Slope | USGS 3DEP 15m DTM



E02 — 46.3003°N 120.1336°W | TPI=-5.4m ▼ VALLEY FLOOR | 18.95km² channel | score=1.000 ✓ McBones Validated



E21 — 47.5086°N 121.1892°W | TPI=-250.1m ▼ VALLEY FLOOR | Most enclosed basin in dataset | score=0.478



E19 — 42.4669°N 117.9392°W | TPI=-3.2m ■ LOW TERRACE | 32.39km² channel area — largest in dataset | score=0.616

Each map shows: LEFT hillshade with blue channel zones · CENTER TPI (blue=valley, red=ridge) · RIGHT slope (dark=flat)

MODEL SUMMARY

Metric	WEST RF	EAST RF v3	v2 East	v1
Points	35	40	17	78
Data	PBDB+iDigBio	PBDB 5-state	PBDB only	PBDB only
CV AUC	0.890±0.105	0.846±0.073	0.566±0.121	0.853±0.063
Delta	**+0.037 vs v1**	**+0.280 vs v2**	–0.287	ref
Confidence	★ HIGH	◆ IMPROVED	■ retired	ref
Validation	—	✓ E02=Coyote Canyon	✗	—
LiDAR TPI	—	**13/20 valley floor**	—	—

TARGET SPECIES

Species	Common Name	Region	Count
Mammuthus columbi	Columbian Mammoth	WA,OR,NV,ID	~12
Mammuthus primigenius	Woolly Mammoth	WA,OR	~5
Mammut americanum	American Mastodon	WA,OR	5
Equus sp.	Pleistocene Horse	WA,OR,NV,MT	22
Camelops hesternus	Yesterday's Camel	WA,OR,NV,ID	13
Paramylodon harlani	Harlan's Ground Sloth	OR,NV	8
Bison sp.	Pleistocene Bison	WA,OR,NV,ID	9
Arctodus simus	Short-faced Bear	WA,OR,NV,MT	5
Canis sp.	Pleistocene Wolf/Dog	OR,NV,ID,MT	9
Odocoileus sp.	Pleistocene Deer	OR,ID,MT	8

TOP 10 FIELD TARGETS — WEST

★ HIGH CONFIDENCE | AUC 0.89 | Willamette Valley / Puget Sound | Cyan in KMZ

Ranked by composite score = 80% ML + 20% SGMC lithology. PRPA permit required on federal land.

#1	**45.175278N, 122.841944W** 45° 10' 31.0" N 122° 50' 31.0" W	ML Prob: 0.9954 Elev: 56m Slope: 0.3° **Score: 1.0000**	Lith: 1.00 ★ HIGH CONFIDENCE
#6	**48.578056N, 122.258611W** 48° 34' 41.0" N 122° 15' 31.0" W	ML Prob: 0.9906 Elev: 53m Slope: 0.2° **Score: 0.9863**	Lith: 1.00 ★ HIGH CONFIDENCE
#7	**45.828056N, 122.605833W** 45° 49' 41.0" N 122° 36' 21.0" W	ML Prob: 0.9897 Elev: 59m Slope: 0.3° **Score: 0.9837**	Lith: 1.00 ★ HIGH CONFIDENCE
#8	**47.466944N, 122.091944W** 47° 28' 1.0" N 122° 5' 31.0" W	ML Prob: 0.9896 Elev: 53m Slope: 1.3° **Score: 0.9834**	Lith: 1.00 ★ HIGH CONFIDENCE
#9	**48.133611N, 123.536389W** 48° 8' 1.0" N 123° 32' 11.0" W	ML Prob: 0.9863 Elev: 56m Slope: 0.7° **Score: 0.9739**	Lith: 1.00 ★ HIGH CONFIDENCE
#10	**48.494722N, 122.897500W** 48° 29' 41.0" N 122° 53' 51.0" W	ML Prob: 0.9734 Elev: 68m Slope: 1.1° **Score: 0.9370**	Lith: 1.00 ★ HIGH CONFIDENCE
#11	**47.647500N, 121.897500W** 47° 38' 51.0" N 121° 53' 51.0" W	ML Prob: 0.9691 Elev: 36m Slope: 0.5° **Score: 0.9246**	Lith: 1.00 ★ HIGH CONFIDENCE
#12	**45.383611N, 122.494722W** 45° 23' 1.0" N 122° 29' 41.0" W	ML Prob: 0.9615 Elev: 56m Slope: 0.1° **Score: 0.9028**	Lith: 1.00 ★ HIGH CONFIDENCE
#13	**48.564167N, 123.064167W** 48° 33' 51.0" N 123° 3' 51.0" W	ML Prob: 0.9593 Elev: 51m Slope: 1.5° **Score: 0.8965**	Lith: 1.00 ★ HIGH CONFIDENCE
#14	**45.661389N, 122.480833W** 45° 39' 41.0" N 122° 28' 51.0" W	ML Prob: 0.9530 Elev: 61m Slope: 0.3° **Score: 0.8786**	Lith: 1.00 ★ HIGH CONFIDENCE

TOP 10 FIELD TARGETS — EAST

◆ IMPROVED | AUC 0.846 (+0.280) | Columbia Basin / Yakima / Basin & Range | Yellow in KMZ

◆ E02 VALIDATED ✓ McBones Coyote Canyon — LiDAR TPI -5.4m — 18.95km² channel zone. E03/E04/E20/E21 in deeper basins (TPI -86m to -250m). E32/E35 deprioritized (ridgetop).

#2	**46.300278N, 120.133611W** 46° 18' 1.0" N 120° 8' 1.0" W	ML Prob: 0.8038 Elev: 205m Slope: 0.0° **Score: 1.0000** **TPI=-5.4m VALLEY FLOOR**	Lith: 1.00 Channel: 18.9km² **◆ E02 VALIDATED ✓**
#3	**46.675278N, 117.758611W** 46° 40' 31.0" N 117° 45' 31.0" W	ML Prob: 0.8038 Elev: 196m Slope: 0.0° **Score: 1.0000** **TPI=-86.3m VALLEY FLOOR**	Lith: 1.00 Channel: 5.3km² **◆ IMPROVED (AUC 0.846)**
#4	**46.633611N, 117.383611W** 46° 38' 1.0" N 117° 23' 1.0" W	ML Prob: 0.8023 Elev: 225m Slope: 0.0° **Score: 0.9941** **TPI=-106.2m VALLEY FLOOR**	Lith: 1.00 Channel: 5.1km² **◆ IMPROVED (AUC 0.846)**
#5	**46.355833N, 120.536389W** 46° 21' 21.0" N 120° 32' 11.0" W	ML Prob: 0.8023 Elev: 242m Slope: 0.0° **Score: 0.9941** **TPI=+0.1m FLAT PLAIN**	Lith: 1.00 Channel: 16.3km² **◆ IMPROVED (AUC 0.846)**
#15	**47.953056N, 117.800278W** 47° 57' 11.0" N 117° 48' 1.0" W	ML Prob: 0.7584 Elev: 571m Slope: 0.0° **Score: 0.8141** **TPI=-15.6m VALLEY FLOOR**	Lith: 1.00 Channel: 16.1km² **◆ IMPROVED (AUC 0.846)**
#16	**47.925278N, 117.355833W** 47° 55' 31.0" N 117° 21' 21.0" W	ML Prob: 0.7584 Elev: 569m Slope: 0.0° **Score: 0.8141** **TPI=-6.7m VALLEY FLOOR**	Lith: 1.00 Channel: 9.6km² **◆ IMPROVED (AUC 0.846)**
#18	**42.689167N, 120.536389W** 42° 41' 21.0" N 120° 32' 11.0" W	ML Prob: 0.7199 Elev: 1325m Slope: 0.9° **Score: 0.6565** **TPI=-15.3m VALLEY FLOOR**	Lith: 1.00 Channel: 20.1km² **◆ IMPROVED (AUC 0.846)**
#19	**42.466944N, 117.939167W** 42° 28' 1.0" N 117° 56' 21.0" W	ML Prob: 0.7101 Elev: 1310m Slope: 1.0° **Score: 0.6165** **TPI=-3.2m LOW TERRACE**	Lith: 1.00 Channel: 32.4km² **◆ IMPROVED (AUC 0.846)**

#20

44.550278N, 117.425278W

44° 33' 1.0" N

117° 25' 31.0" W

ML Prob: 0.6836

Elev: 780m Slope: 0.0°

Score: 0.5078

TPI=-97.9m VALLEY FLOOR

Lith: 1.00

Channel: 3.4km²

◆ IMPROVED (AUC 0.846)

#21

47.508611N, 121.189167W

47° 30' 31.0" N

121° 11' 21.0" W

ML Prob: 0.6764

Elev: 904m Slope: 0.0°

Score: 0.4785

TPI=-250.1m VALLEY FLOOR

Lith: 1.00

Channel: 1.9km²

◆ IMPROVED (AUC 0.846)

FIELD PREPARATION NOTES

Primary West — Tualatin Valley / Puget Sound (★)

W01-W05 in Tualatin Valley and Willamette Valley. Low-elevation alluvial basins, lith_score=1.0. Access via I-5. Verify ownership — mostly agricultural. Puget Sound sites near Snohomish/Skagit drainages have documented Pleistocene records.

Primary East — Yakima/Kennewick Corridor (◆ VALIDATED)

E02 near Kennewick confirmed productive by McBones Coyote Canyon. TPI=-5.4m, 18.95km² channel zone. E03/E04 in Palouse at TPI -86m/-106m are in deeper basins — same Missoula Flood corridor. These are buried sites — monitor for gravel pit operations, road construction, irrigation work in the 800-1200ft band. Contact McBones (mcbones.org) to volunteer at the active Kennewick dig.

High-Priority Deep Basins (◆)

E20 (TPI -98m, Baker City OR corridor) and E21 (TPI -250m, Wenatchee/Entiat) are the most topographically enclosed targets in the dataset. E21 especially — a -250m TPI basin at 47.5N is unusual and warrants closer investigation. Check satellite imagery for exposed cut banks or active erosion.

SE Oregon / Basin & Range (◆)

E18/E19/E23/E27 target SE Oregon basin-and-range playas. E19 has the largest channel zone in the dataset at 32.39km² — broad slack-water basin in the Owyhee drainage. Actively eroding terrain, surface prospecting feasible. Access via US-395. High-clearance 4WD required. Nearest services 50+ miles.

Deprioritized Targets — DO NOT FIELD

E29 (TPI +4.6m), E31 (+1.4m), E32 (+40m), E35 (+13m) are ridge/slope positions confirmed by LiDAR TPI. ML probability was elevated but terrain geometry is wrong for fossil burial. Remove from field planning.

Permits

BLM OR/WA: 503-808-6002 · BLM NV: 775-861-6400 · BLM ID: 208-373-4000 · BLM MT: 406-896-5000. PRPA permit required for all vertebrate collection on federal land. Private land needs written permission.

QUICK REFERENCE — WEST (★ AUC 0.890)

#	Decimal Degrees	DMS Lat	DMS Lon	Score
W01	45.175278, -122.841944	45° 10' 31.0" N	122° 50' 31.0" W	1.0000
W06	48.578056, -122.258611	48° 34' 41.0" N	122° 15' 31.0" W	0.9863
W07	45.828056, -122.605833	45° 49' 41.0" N	122° 36' 21.0" W	0.9837
W08	47.466944, -122.091944	47° 28' 1.0" N	122° 5' 31.0" W	0.9834
W09	48.133611, -123.536389	48° 8' 1.0" N	123° 32' 11.0" W	0.9739
W10	48.494722, -122.897500	48° 29' 41.0" N	122° 53' 51.0" W	0.9370
W11	47.647500, -121.897500	47° 38' 51.0" N	121° 53' 51.0" W	0.9246
W12	45.383611, -122.494722	45° 23' 1.0" N	122° 29' 41.0" W	0.9028
W13	48.564167, -123.064167	48° 33' 51.0" N	123° 3' 51.0" W	0.8965
W14	45.661389, -122.480833	45° 39' 41.0" N	122° 28' 51.0" W	0.8786
W17	44.480833, -122.980833	44° 28' 51.0" N	122° 58' 51.0" W	0.7664
W24	47.008611, -122.897500	47° 0' 31.0" N	122° 53' 51.0" W	0.4587
W28	42.994722, -123.314167	42° 59' 41.0" N	123° 18' 51.0" W	0.3114
W36	44.008611, -122.022500	44° 0' 31.0" N	122° 1' 21.0" W	0.1152
W37	42.272500, -123.591944	42° 16' 21.0" N	123° 35' 31.0" W	0.0744
W39	42.439167, -123.369722	42° 26' 21.0" N	123° 22' 11.0" W	0.0500
W41	42.814167, -123.883611	42° 48' 51.0" N	123° 53' 1.0" W	0.0000

QUICK REFERENCE — EAST (◆ AUC 0.846)

#	Decimal Degrees	DMS Lat	DMS Lon	Score	TPI	Tier
E02	46.300278, -120.133611	46° 18' 1.0" N	120° 8' 1.0" W	1.0000	-5m	1A
E03	46.675278, -117.758611	46° 40' 31.0" N	117° 45' 31.0" W	1.0000	-86m	1B
E04	46.633611, -117.383611	46° 38' 1.0" N	117° 23' 1.0" W	0.9941	-106m	1B
E05	46.355833, -120.536389	46° 21' 21.0" N	120° 32' 11.0" W	0.9941	+0m	2
E15	47.953056, -117.800278	47° 57' 11.0" N	117° 48' 1.0" W	0.8141	-16m	1A
E16	47.925278, -117.355833	47° 55' 31.0" N	117° 21' 21.0" W	0.8141	-7m	1B
E18	42.689167, -120.536389	42° 41' 21.0" N	120° 32' 11.0" W	0.6565	-15m	1A
E19	42.466944, -117.939167	42° 28' 1.0" N	117° 56' 21.0" W	0.6165	-3m	1A
E20	44.550278, -117.425278	44° 33' 1.0" N	117° 25' 31.0" W	0.5078	-98m	1B
E21	47.508611, -121.189167	47° 30' 31.0" N	121° 11' 21.0" W	0.4785	-250m	1B
E22	43.647500, -120.189167	43° 38' 51.0" N	120° 11' 21.0" W	0.4784	—	?
E23	43.286389, -117.605833	43° 17' 11.0" N	117° 36' 21.0" W	0.4784	-21m	1B
E25	44.814167, -117.841944	44° 48' 51.0" N	117° 50' 31.0" W	0.4410	+1m	2
E26	46.911389, -120.328056	46° 54' 41.0" N	120° 19' 41.0" W	0.3705	-16m	2

E27	43.091944, -117.091944	43° 5' 31.0" N	117° 5' 31.0" W	0.3673	-14m	1A
E29	42.161389, -120.453056	42° 9' 41.0" N	120° 27' 11.0" W	0.2929	+5m	X
E30	46.078056, -118.925278	46° 4' 41.0" N	118° 55' 31.0" W	0.2774	—	?
E31	46.244722, -118.008611	46° 14' 41.0" N	118° 0' 31.0" W	0.2714	+1m	X
E32	46.383611, -117.189167	46° 23' 1.0" N	117° 11' 21.0" W	0.2524	+40m	X
E33	46.536389, -120.800278	46° 32' 11.0" N	120° 48' 1.0" W	0.2510	—	?
E34	42.661389, -120.425278	42° 39' 41.0" N	120° 25' 31.0" W	0.2300	-8m	1A
E35	46.911389, -118.550278	46° 54' 41.0" N	118° 33' 1.0" W	0.1345	+13m	X
E38	42.175278, -120.550278	42° 10' 31.0" N	120° 33' 1.0" W	0.0664	—	?
E40	43.855833, -120.841944	43° 51' 21.0" N	120° 50' 31.0" W	0.0324	-4m	1A
E42	43.703056, -117.091944	43° 42' 11.0" N	117° 5' 31.0" W	0.0000	—	?

Project IceWave v4 | West AUC 0.89 ★ | East AUC 0.846 ◆ | E02 Validated ✓ | LiDAR TPI 13/20 Valley Floor | March 2026 | github.com/bdgroves/project_ice_wave